AZURE MACHINE LEARNING DESIGNER PROJECT REPORT

Sales Data Prediction using Linear Regression

Introduction:

This course project demonstrates the development of a machine learning pipeline using **Microsoft Azure Machine Learning Designer**. The objective was to build a regression model that predicts the **Quantity Ordered (QUANTITYORDERED)** based on historical sales data.

The project was completed using Azure's drag-and-drop Designer interface, allowing the entire machine learning workflow to be created without writing code. The pipeline includes data exploration, feature selection, data splitting, model training, prediction, and evaluation.

Dataset Overview:

The dataset contains **2,823 sales records** with **25 attributes** related to customer orders. It includes information such as:

- Order Number
- Quantity Ordered (Target Variable)
- Price Each
- Sales
- Order Date
- Product Line
- Customer Name
- Country
- Deal Size
- Status
- Product Code

The target variable selected for prediction was **QUANTITYORDERED**.

Pipeline Components:

The Azure ML Designer pipeline consists of the following components:

1. Summarize Data

This component provides a statistical overview of the dataset, helping understand data distribution, missing values, and feature characteristics before model training.

2. Select Columns in Dataset

Only the relevant features required for training were selected, while unnecessary columns were excluded to simplify the model.

3. Split Data

The dataset was divided into training and testing datasets. The training set was used to build the model, while the testing set was reserved for evaluating its performance.

4. Linear Regression

A Linear Regression algorithm was selected because the target variable (Quantity Ordered) is numerical and continuous.

5. Train Model

The algorithm learned the relationship between the selected features and the target variable using the training dataset.

6. Score Model

The trained model generated predictions for the test dataset.

7. Evaluate Model

Finally, Azure Machine Learning calculated several evaluation metrics to measure the prediction accuracy of the model.

Model Evaluation:

The model produced the following evaluation results:

Metric	Value	Interpretation
Coefficient of Determination (R^2)	**0.6505**	The model explains approximately **65% of the variation** in Quantity Ordered.
Mean Absolute Error (MAE)	**4.3808**	On average, predictions differ from actual values by about **4.38 units**.
Root Mean Squared Error (RMSE)	**5.6550**	Indicates the average prediction error while giving more weight to larger errors. Lower values indicate better performance.
Relative Absolute Error (RAE)	**0.5406**	The model performs significantly better than a simple baseline prediction.
Relative Squared Error (RSE)	**0.3495**	Shows a substantial reduction in prediction error compared with a baseline model.

Overall Performance:

The Linear Regression model achieved an **R^2 score of 0.65**, indicating that it captures a significant portion of the relationship between the input features and the target variable. While there is still room for improvement through feature engineering or advanced algorithms, the model provides a solid baseline and demonstrates effective use of Azure Machine Learning Designer.

Conclusion:

This project provided practical experience in designing an end-to-end machine learning workflow using Azure Machine Learning Designer. It covered the complete ML lifecycle, including data exploration, preprocessing, model training, prediction, and evaluation without requiring manual coding.

The project also highlighted the importance of selecting appropriate features, splitting data correctly, and evaluating model performance using multiple metrics rather than relying on a single score.

Overall, this project strengthened my understanding of Azure Machine Learning, regression modeling, and cloud-based AI development.